Chapter 12 — Nitrogen and Sulfur

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 280–282

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Nitrogen does not readily participate in combustion because:

(a) It is denser than oxygen.
(b) It has a high specific heat capacity.
(c) Breaking the N≡N bond requires a large amount of energy.
(d) It is a noble gas.

Answer: (c) Breaking the N≡N bond requires a large amount of energy. — The N≡N triple bond is very strong (~944 kJ mol^{-1}), making N_2 unreactive.

II. Heating an ammonium salt with strong alkali gives a pungent gas that turns damp red litmus blue. The gas is:

(a) Hydrogen (H_2)
(b) Carbon dioxide (CO_2)
(c) Ammonia (NH_3)
(d) Sulfur dioxide (SO_2)

Answer: (c) Ammonia (NH_3) — $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$; ammonia is pungent and basic.

III. The shape of ammonium is:

(a) Pyramidal
(b) Triangular planar
(c) Tetrahedral
(d) Linear

Answer: (c) Tetrahedral — NH_4^+ has four bond pairs and no lone pair → tetrahedral.

IV. In a catalytic converter, conversion of NOx into N_2 and O_2 is a process of:

(a) Oxidation
(b) Reduction
(c) Combustion
(d) Neutralization

Answer: (b) Reduction — Nitrogen in NOx is reduced (oxidation number decreases) to N_2.

V. PAN formation starts when _____ reacts with the hydrocarbon.

(a) NO
(b) NO_2
(c) O_3
(d) HO•

Answer: (c) O_3 — According to the textbook, ozone oxidises the hydrocarbon to an aldehyde, starting the PAN-forming sequence.

The mechanism also involves HO• radicals; the question wording is simplified.

VI. Nitrification is the process by which:

(a) Atmospheric nitrogen is converted into ammonia.
(b) Nitrate is converted into nitrogen gas.
(c) Ammonia is converted into nitrite and then nitrate.
(d) Organic nitrogen is converted into ammonia.

Answer: (c) Ammonia is converted into nitrite and then nitrate. — Nitrification is the bacterial oxidation $NH_3 \rightarrow NO_2^- \rightarrow NO_3^-$.

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VII. The most stable species in an acidic environment is:

(a) SO_4^{2-} **(b)** SO_2
(c) H_2S **(d)** S

Answer: (a) SO_4^{2-} — The textbook notes that in acidic environment SO_4^{2-} is kinetically stable due to strong O–S bonds.

The same section also states that reduced forms (H_2S) are favoured under acidic conditions, so the wording is ambiguous; (a) follows the "kinetically stable" statement.

VIII. Which gas is used in separating hard water from normal water?

(a) SO_2 **(b)** H_2S
(c) NH_3 **(d)** NO_2

Answer: (c) NH_3 — Ammonia (an alkali) can precipitate hardness-causing ions such as Mg^{2+}, helping to distinguish/soften hard water.

The wording is unclear; hard/soft water is usually distinguished with soap, not a gas.

IX. The oxidation state of sulfur in H_2SO_4 is:

(a) +1 **(b)** +2
(c) +4 **(d)** +6

Answer: (d) +6 — $2(+1) + x + 4(-2) = 0 \rightarrow x = +6$.

X. The drug omeprazole contains _____ group.

(a) Thiol **(b)** Sulfoxide
(c) Bisulfide **(d)** Sulfone

Answer: (b) Sulfoxide — Omeprazole contains a sulfoxide (S=O) group.

XI. SO_2 can be oxidized to SO_3 in the presence of:

(a) Catalyst (e.g. vanadium(V) oxide) and high temperature **(b)** Catalyst (e.g. iron) and low temperature
(c) Strong reducing agent and high pressure **(d)** Dilute acid and room temperature

Answer: (a) Catalyst (e.g. vanadium(V) oxide) and high temperature — The Contact process uses V_2O_5 at about 450 °C.

XII. SO_3 is not directly dissolved in water in the Contact process because the reaction is:

(a) Too slow **(b)** Reversible and would result in a low yield
(c) Highly exothermic and produces a mist of sulfuric acid **(d)** Requires very high pressures

Answer: (c) Highly exothermic and produces a mist of sulfuric acid — Direct dissolution is highly exothermic and forms a difficult-to-condense acid mist; SO_3 is absorbed in H_2SO_4 to form oleum instead.

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Q.2 — Short Answer Questions

a. List two reasons for the inertness of N_2.

(i) The N≡N triple bond has very high bond energy (~944 kJ mol^{-1}). (ii) N_2 is non-polar with a stable electronic configuration, so it has little tendency to react.

b. How is nitrogen isolated from air?

Air is liquefied and then fractionally distilled. Nitrogen (b.p. −196 °C) distils off before oxygen (b.p. −183 °C), giving nitrogen gas.

c. Why is ammonia (NH_3) a weak base?

NH_3 accepts a proton via its lone pair, but it ionizes only partially in water: $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$. The equilibrium lies far to the left, so it is a weak base.

d. Write down the reactions of photochemical smog formation.

$N_2 + O_2 \rightarrow 2NO$ (in engines); $2NO + O_2 \rightarrow 2NO_2$; NO_2 + sunlight → NO + O; $O + O_2 \rightarrow O_3$. Ozone and NOx then react with hydrocarbons to form aldehydes and PAN.

e. What is the construction and function of a catalytic converter?

It is a honeycomb ceramic/metal structure coated with Pt, Pd and Rh fitted in the exhaust. It catalyses conversion of CO, NOx and unburnt hydrocarbons into CO_2, N_2 and H_2O.

f. Why is sulfur quite unreactive at room temperature?

Sulfur exists as stable S_8 rings with strong S–S bonds; at room temperature there is insufficient energy to break these bonds, so it is relatively unreactive.

g. Which are the most stable oxidation states of sulfur in water at pH = 0 and pH = 14?

At pH = 0 (acidic) reduced forms such as H_2S (−2) are favoured; at pH = 14 (basic) oxidized forms such as SO_4^{2-} (+6) are more stable. (The textbook also notes SO_4^{2-} is kinetically stable in acid.)

h. How does sulfur react with halogens?

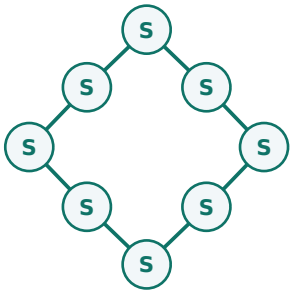
With fluorine: $S + 3F_2 \rightarrow SF_6$ (and SF_4). With chlorine: $2S + Cl_2 \rightarrow S_2Cl_2$; $S_2Cl_2 + Cl_2 \rightarrow 2SCl_2$.

i. Draw the structures of cyclo-octasulfur (S_8) and sulfuric acid.

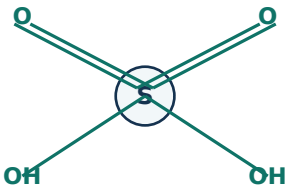
S_8 is a puckered crown-shaped ring of eight S atoms. H_2SO_4 has a central S bonded to two =O and two –OH groups (tetrahedral around S).

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Cyclo-octasulfur, S_8 (crown)



Sulfuric acid, H_2SO_4



j. What is the role of sulfur in the vulcanization of rubber?

Sulfur forms cross-links (disulfide bridges) between rubber polymer chains, improving elasticity, strength, hardness and resistance to heat and solvents.

k. What is the composition and the chemical reaction of gunpowder combustion?

Gunpowder (black powder) is a mixture of KNO_3, charcoal (C) and sulfur. A representative reaction: $2KNO_3 + S + 3C \rightarrow K_2S + N_2 + 3CO_2$.

l. What is the importance of disulfide bridges?

Disulfide (–S–S–) bridges cross-link protein chains (e.g. keratin), stabilising the three-dimensional structure and giving strength and rigidity.

m. Write the self-ionization equation of sulfuric acid and its ionization in water.

Self-ionization: $2H_2SO_4 \rightleftharpoons H_3SO_4^+ + HSO_4^-$. In water: $H_2SO_4 \rightarrow H^+ + HSO_4^-$ (first, complete); $HSO_4^- \rightleftharpoons H^+ + SO^{2-}$ (second, partial).

n. Give two examples where sulfuric acid acts as a dehydrating agent.

(i) Charring sugar: $C_{12}H_{22}O_{11} \rightarrow$(conc. H_2SO_4) $12C + 11H_2O$. (ii) Dehydration of ethanol: $C_2H_5OH \rightarrow$(conc. H_2SO_4) $C_2H_4 + H_2O$.

o. How does V_2O_5 catalyze the formation of SO_3?

V_2O_5 provides an alternative pathway: it oxidizes SO_2 to SO_3 while being reduced to V_2O_4, then is re-oxidized by O_2 back to V_2O_5, lowering the activation energy.

p. What is the purpose of formation of oleum?

SO_3 is absorbed in concentrated H_2SO_4 to form oleum ($H_2S_2O_7$) to avoid the highly exothermic direct reaction with water that produces an acid mist. Oleum is then carefully diluted to give sulfuric acid.

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Descriptive / Long Questions

Q.3 Explain the preparation and basicity of ammonia.

Laboratory preparation: heating an ammonium salt with a strong base: $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$. The gas is dried over quicklime (CaO) and collected by upward displacement of air.

Industrial preparation (Haber process): $N_2 + 3H_2 \rightleftharpoons 2NH_3$, at ~450 °C, ~200 atm, Fe catalyst.

Basicity: NH_3 is a weak Bronsted-Lowry/Lewis base. The lone pair on N accepts a proton: $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$. It ionizes only partially, so it is a weak base.

Q.4 How do oxides of nitrogen (NOx) cause the formation of photochemical smog and PAN? Give the mechanism.

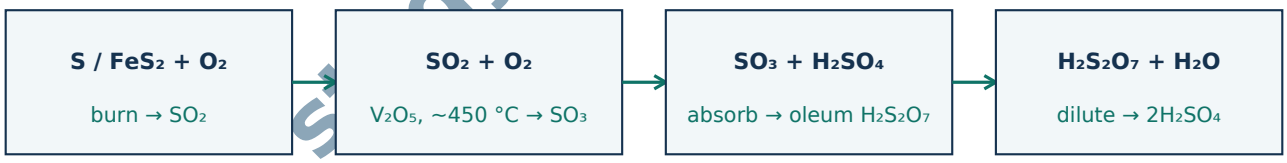
NOx and VOCs in sunlight form photochemical smog. NO_2 photolyses: $NO_2 + h\nu \rightarrow NO + O$; $O + O_2 \rightarrow O_3$. Ozone oxidises hydrocarbons to aldehydes.

PAN mechanism: aldehyde + HO• → acyl radical (RCO•); RCO• + O_2 → peroxyacyl radical (RCO_3•); RCO_3• + NO_2 → peroxyacyl nitrate (PAN, RCO_3NO_2). PAN and ozone are harmful oxidants in smog.

Q.5 Give a flowsheet diagram and equations involved in the Contact process.

Steps: (1) $S + O_2 \rightarrow SO_2$ (or $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$). (2) $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (V_2O_5, ~450 °C, 1–2 atm). (3) $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$ (oleum). (4) $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$.

Contact process — flowsheet



SO_3 is not dissolved directly in water (acid mist); oleum is diluted carefully.

Q.6 Discuss sulfuric acid as an oxidizing agent and a dehydrating agent with three reactions for each.

Oxidizing agent (hot, concentrated): $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$; $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$; $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$.

Dehydrating agent: $C_{12}H_{22}O_{11} \rightarrow 12C + 11H_2O$ (sugar charring); $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (ethanol to ethene); $HCOOH \rightarrow CO + H_2O$ (methanoic acid to CO).

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Quick Check Solutions (in-chapter practice boxes)

Quick Check 12.1 (book p. 267)

a. Why N_2 gas is used in food packaging.

N_2 is inert and excludes oxygen, preventing oxidation/rancidity and microbial growth, so food stays fresh.

b. Both CO and N_2 have triple bonds. Why is CO more reactive than N_2?

N_2 is non-polar with a very strong triple bond; CO is polar and has a lone pair on carbon, making it more reactive (e.g. it binds to hemoglobin and acts as a ligand).

c. Why a farmer should not apply ammonium fertilizer with lime; give reactions.

Lime ($Ca(OH)_2$) reacts with ammonium salts to release NH_3 gas, losing nitrogen: $(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2NH_3\uparrow + 2H_2O$.

Quick Check 12.2 (book p. 270)

a. Draw structures and explain bonding of (i) N_2O (ii) NO (iii) NO_2.

(i) N_2O: linear N≡N–O (resonance N=N=O); covalent. (ii) NO: diatomic with an unpaired electron (radical), bond order ~2.5. (iii) NO_2: bent molecule with an unpaired electron on N; resonance between the two N–O bonds.

b. What does PAN stand for? Give its general formula.

Peroxyacyl nitrate; general formula RCO_3NO_2 (e.g. $CH_3CO_3NO_2$).

c. Formulas of compounds responsible for PAN formation.

Hydrocarbons/VOCs (e.g. CH_3CHO), NO_2 and O_3 (and HO• radicals).

d. Mg burns in N_2O; how does the product confirm the structure of N_2O?

$Mg + N_2O \rightarrow MgO + N_2$. The formation of N_2 gas shows that N_2O contains a nitrogen–nitrogen unit that is released as N_2, consistent with the N≡N–O structure.

Quick Check 12.3 (book p. 272)

a. Write reduction and oxidation reactions in a catalytic converter.

Reduction: $2NOx \rightarrow N_2 + xO_2$ (e.g. $2NO \rightarrow N_2 + O_2$; $2NO_2 \rightarrow N_2 + 2O_2$). Oxidation: $2CO + O_2 \rightarrow 2CO_2$; hydrocarbons + $O_2 \rightarrow CO_2 + H_2O$.

b. Basic principle and role of the catalyst.

The converter passes exhaust over Pt/Pd/Rh catalysts, which lower the activation energy for redox reactions, converting CO, NOx and unburnt hydrocarbons into CO_2, N_2 and H_2O.

c. Do hybrid and electric cars have catalytic converters?

Hybrid cars yes (they have an internal-combustion engine and exhaust). Pure electric cars no (no combustion engine/exhaust).

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Quick Check 12.4 (book p. 274)

a. Oxidation state of S in (i) SO_3 (ii) H_2SO_4 (iii) SO_4^{2-} (iv) $S_2O_3^{2-}$.

(i) +6. (ii) +6. (iii) +6. (iv) Average +2 (the two S atoms are in different environments, −1 and +5).

b. Most common oxidation states of sulfur and why.

−2, +4 and +6. S has 6 valence electrons; it can gain 2 (−2) or share/use 4 or 6 electrons with more electronegative elements (+4, +6).

c. Involvement of d orbitals in variable oxidation states of S.

Sulfur (Period 3) has vacant 3d orbitals, so it can expand its octet and use 4 or 6 electrons in bonding, giving +4 and +6 states.

Quick Check 12.5 (book p. 275)

a. Function of SO_2 and sulfite salts in preserving food.

They act as antioxidants and antimicrobial agents, preventing oxidation and microbial spoilage (e.g. in dried fruits and wines).

b. Reaction of S with HCl and NaOH.

S does not react with HCl. With hot concentrated NaOH it disproportionates: $3S + 6NaOH \rightarrow 2Na_2S + Na_2SO_3 + 3H_2O$.

Quick Check 12.6 (book p. 277)

a. Why high temperature and catalyst are needed to form SO_3.

The oxidation of SO_2 has a high activation energy; a catalyst (V_2O_5) lowers it and a high temperature (~450 °C) gives a useful rate while maintaining a reasonable yield.

b. Dehydration reactions of conc. H_2SO_4 with starch and oxalic acid.

Starch: $(C_6H_{10}O_5)_n \rightarrow$(conc. H_2SO_4) $6nC + 5nH_2O$ (charring). Oxalic acid: $H_2C_2O_4 \rightarrow$(conc. H_2SO_4) $CO + CO_2 + H_2O$.

c. How V_2O_5 functions in converting SO_2 to SO_3.

V_2O_5 oxidizes SO_2 to SO_3 and is reduced to V_2O_4; O_2 then re-oxidizes V_2O_4 back to V_2O_5, so the catalyst is regenerated.

Quick Check 12.7 (book p. 279)

a. Dehydration reactions of conc. H_2SO_4 with starch and oxalic acid.

Starch → carbon + water (charring). Oxalic acid → $CO + CO_2 + H_2O$.

b. How conc. H_2SO_4 reacts with NaCl; importance.

$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl\uparrow$. This is the laboratory preparation of hydrogen chloride gas.

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c. Difference between oxidizing power of cold and hot sulfuric acid.

Cold dilute H_2SO_4 is a weak oxidizing agent (mainly acidic); hot concentrated H_2SO_4 is a strong oxidizing agent, oxidizing metals/non-metals and being reduced to SO_2.

Textbook checks flagged in this chapter

• p. 281, MCQ VII: the text gives conflicting statements about stability in acid; the answer follows the "SO_4^{2-} is kinetically stable" statement.

• p. 281, MCQ VIII: "separating hard water from normal water" is unclear wording; hard/soft water is normally distinguished with soap.

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